Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory
Experiment 2
Implementation of a Multi-Layer Perceptron (MLP) for Multi-Class Image Classification

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

M Shakthi Sridevi
24011101063
AI DS A2

1. Objective

Implement an MLP using TensorFlow/Keras on the Fashion-MNIST dataset. Learn image preprocessing, flattening, model construction, training, evaluation, and automated hyperparameter optimization.

2. Background Theory

An MLP consists of an input layer, hidden layers and an output layer.

$$z^{(l)} = W^{(l)}a^{(l-1)} + b^{(l)}, \qquad a^{(l)} = f(z^{(l)})$$

Output layer:

$$\hat{y} = \text{Softmax}(z)$$

Loss:

$$L = -\sum_i y_i \log(\hat{y}_i)$$

Recommended architecture:

$$784 \rightarrow \text{Dense(128, ReLU)} \rightarrow \text{Dense(64, ReLU)} \rightarrow \text{Dense(10, Softmax)}$$

3. Dataset

Dataset: Fashion-MNIST
Training Images: 60,000
Testing Images: 10,000
Classes: 10
Image Size: 28×28

4. Experimental Procedure

Task 1: Dataset Exploration

- Load Fashion-MNIST.
- Print dataset dimensions.
- Display ten sample images.
- Plot class distribution.

Expected Output: Dataset summary and sample images.

Task 2: Data Preprocessing
- Flatten images.
- Normalize pixel values.
- Print tensor shapes before and after preprocessing.

Task 3: Model Construction
Construct an MLP with
- Input layer (784)
- Dense(128, ReLU)
- Dense(64, ReLU)
- Dense(10, Softmax)

Display `model.summary()`.

Task 4: Model Training
Compile using
- Optimizer: Adam
- Loss: Categorical Cross Entropy
- Metric: Accuracy

Train for 20 epochs with batch size 32.

Task 5: Model Evaluation
Compute
- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

5. Source Code

`https://github.com/Shakthii27/Deep-Learning-Lab-2/`

6. Hyperparameter Optimization

Instead of manually selecting hyperparameters, students shall perform automated hyperparameter optimization using either **GridSearchCV** or **RandomizedSearchCV** (recommended) together with the SciKeras wrapper.

Optimize the following search space.

Hyperparameter	Candidate Values
Hidden Layers	1, 2, 3
Hidden Neurons	32, 64, 128, 256
Learning Rate	0.1, 0.01, 0.001
Batch Size	16, 32, 64, 128
Epochs	10, 20, 30
Optimizer	SGD, Adam, RMSProp
Activation Function	ReLU, Tanh, Sigmoid
Dropout Rate (Optional)	0.0, 0.2, 0.5

7. Results

Task 1: Dataset Loading

The Fashion-MNIST dataset was successfully loaded. It contains 60,000 training images and 10,000 testing images. Each image is of size 28×28 pixels.

Table 1: Dataset Dimensions

Dataset	**Shape**
Training Images	(60000, 28, 28)
Training Labels	(60000,)
Testing Images	(10000, 28, 28)
Testing Labels	(10000,)

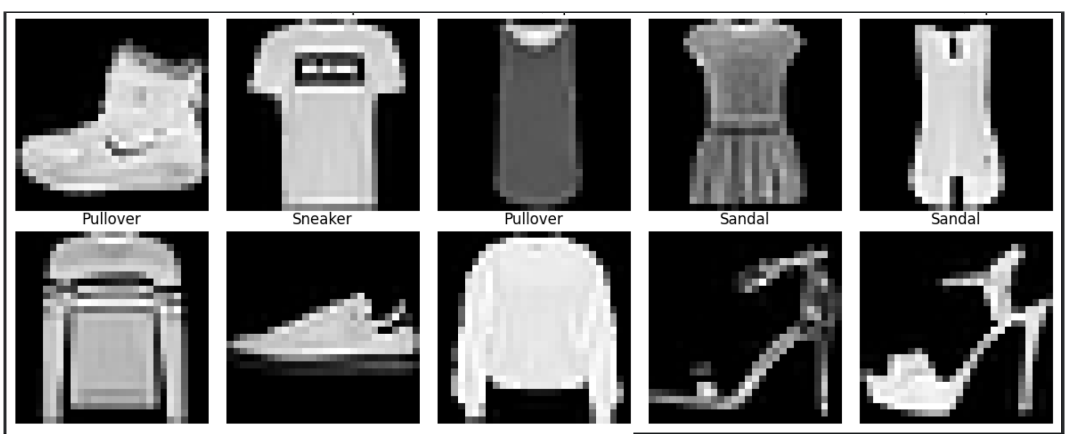

Figure 1: Sample Images from Fashion-MNIST Dataset

Figure 2: Class Distribution of Fashion-MNIST Dataset

Task 2: Data Preprocessing

The images were flattened from 28 × 28 into 784-dimensional vectors and normalized. Labels were converted into one-hot encoded vectors.

Table 2: Dataset Before and After Preprocessing

Dataset	**Before**	**After**
Training Images	(60000,28,28)	(60000,784)
Testing Images	(10000,28,28)	(10000,784)
Training Labels	(60000,)	(60000,10)
Testing Labels	(10000,)	(10000,10)

Task 3: Baseline MLP Model

A Multi-Layer Perceptron (MLP) consisting of two hidden layers was constructed for image classification.

Table 3: Baseline MLP Architecture

Layer	**Output Shape**	**Parameters**
Dense (128 neurons)	(None,128)	100480
Dense (64 neurons)	(None,64)	8256
Dense (10 neurons)	(None,10)	650
Total Parameters		109386
Trainable Parameters		109386
Non-trainable Parameters		0

Task 4: Model Training

The baseline MLP model was trained for 20 epochs using the Adam optimizer. The model achieved a final training accuracy of 93.33% and validation accuracy of 88.72%.

Table 4: Training Summary

Epoch	Train Accuracy	Train Loss	Validation Accuracy	Validation Loss
1	0.8176	0.5155	0.8553	0.4053
2	0.8614	0.3793	0.8688	0.3652
3	0.8744	0.3423	0.8694	0.3631
4	0.8834	0.3162	0.8772	0.3350
5	0.8890	0.2985	0.8827	0.3252
6	0.8958	0.2824	0.8753	0.3368
7	0.8981	0.2698	0.8798	0.3339
8	0.9034	0.2574	0.8824	0.3352
9	0.9068	0.2486	0.8851	0.3271
10	0.9108	0.2368	0.8874	0.3237
11	0.9123	0.2307	0.8868	0.3202
12	0.9149	0.2242	0.8851	0.3348
13	0.9173	0.2170	0.8875	0.3260
14	0.9213	0.2088	0.8806	0.3461
15	0.9224	0.2050	0.8910	0.3367
16	0.9251	0.1979	0.8906	0.3340
17	0.9286	0.1891	0.8842	0.3746
18	0.9288	0.1874	0.8885	0.3496
19	0.9301	0.1804	0.8919	0.3436
20	0.9333	0.1744	0.8872	0.3718

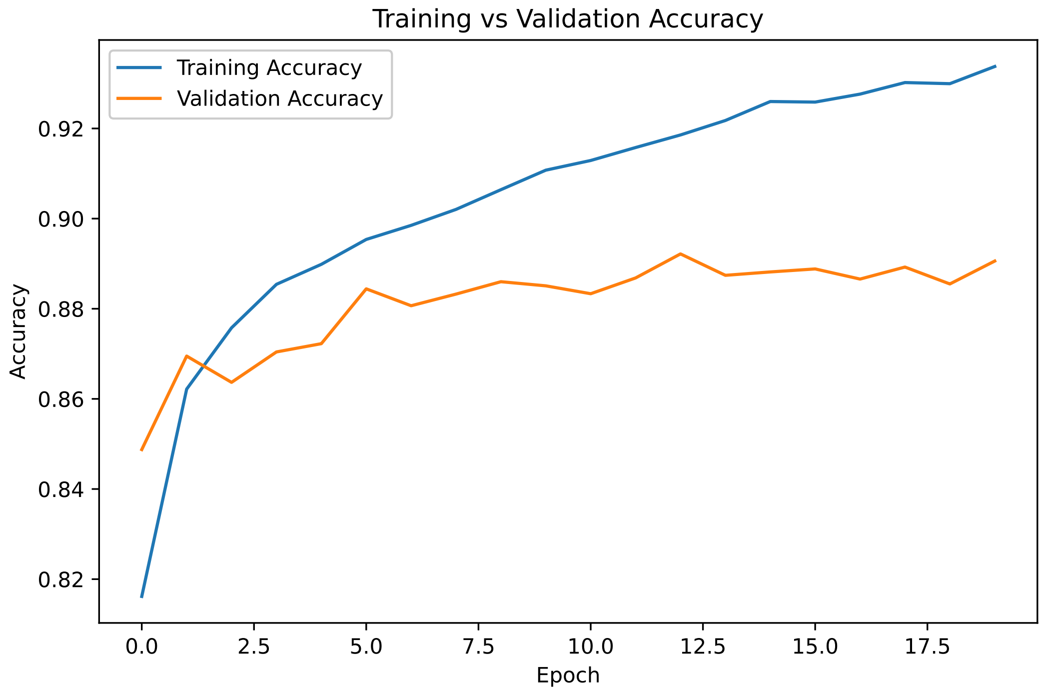

Figure 3: Training and Validation Accuracy

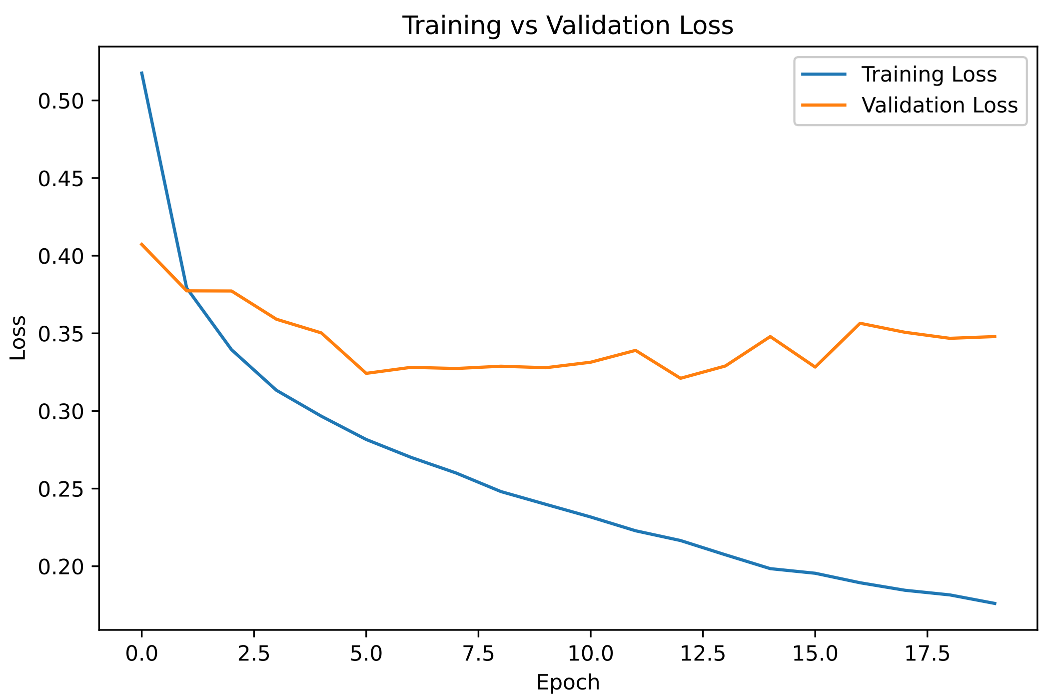

Figure 4: Training and Validation Loss

Task 5: Baseline Model Evaluation

The trained MLP achieved an overall testing accuracy of 88.48%.

Table 5: Performance Metrics

Accuracy	0.8848
Precision	0.8834
Recall	0.8848
F1-Score	0.8833

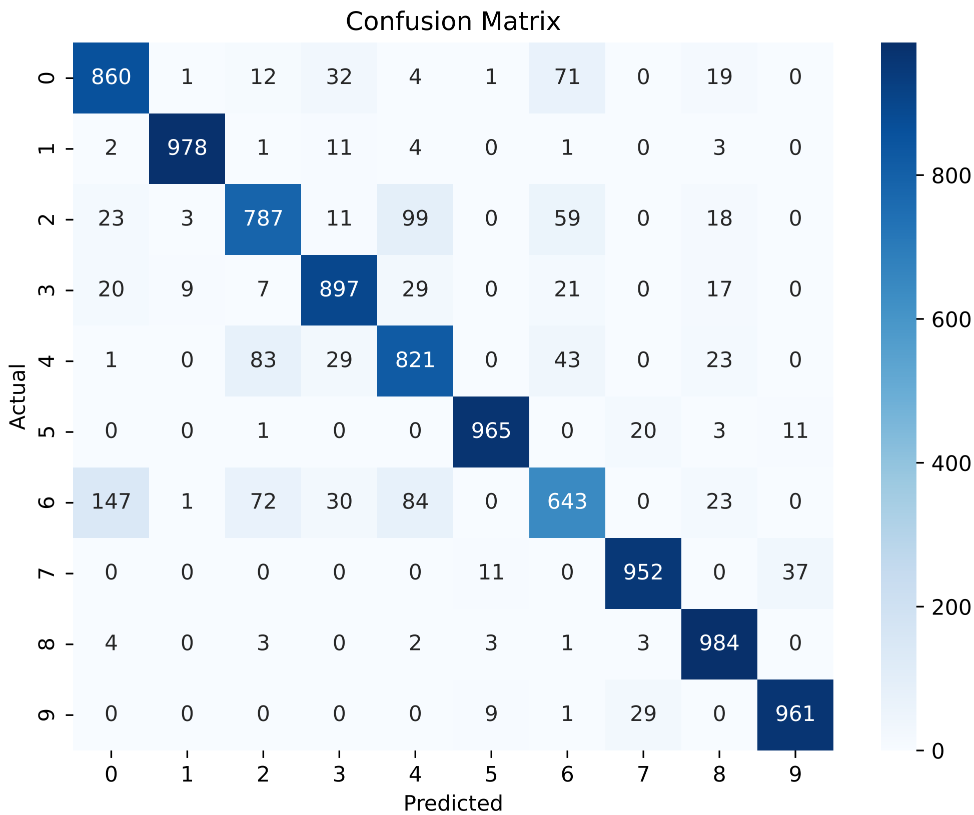

Figure 5: Confusion Matrix

Re-trained Model

Randomized Search Cross Validation was performed to identify the optimal hyperparameters for the MLP model.

Table 6: Best Hyperparameters

Optimizer	Adam
Learning Rate	0.001
Hidden Layers	2
Neurons	64
Activation	ReLU
Dropout Rate	0.0
Epochs	5
Batch Size	64
Best Cross Validation Score	0.8671

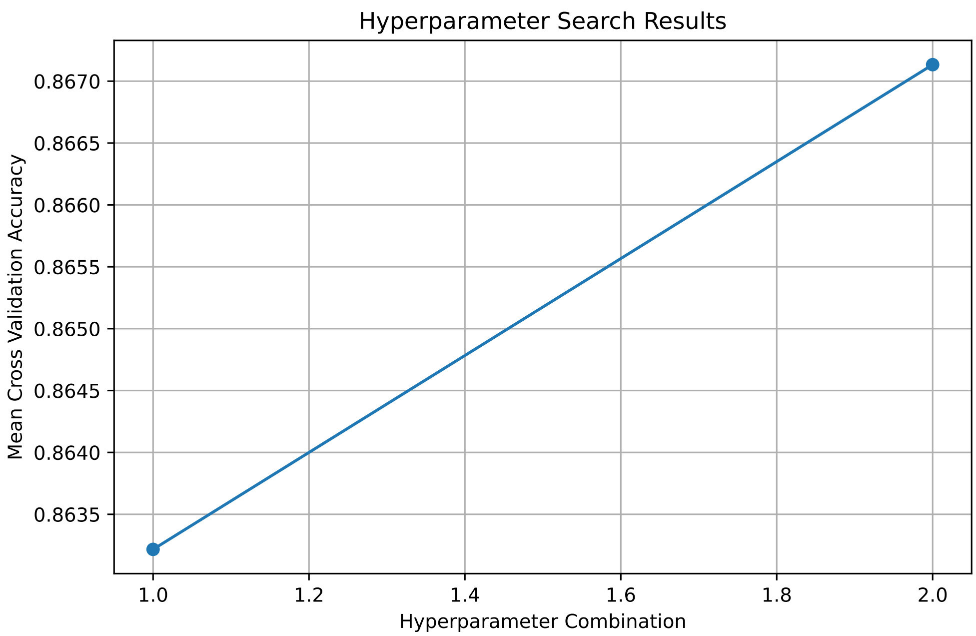

Figure 6: Hyperparameter Search Results

Optimized Model Evaluation

The optimized model was retrained using the best hyperparameters and evaluated on the testing dataset.

Table 7: Optimized Model Performance

Accuracy	0.8701
Precision	0.8718
Recall	0.8701
F1-Score	0.8660

Plots with Inference

Randomized Search Cross Validation (RandomizedSearchCV) was performed to determine the optimal hyperparameters for the Multi-Layer Perceptron (MLP). The search evaluated different combinations of hidden layers while keeping the remaining parameters fixed. The best-performing configuration was selected based on the highest mean cross-validation accuracy.

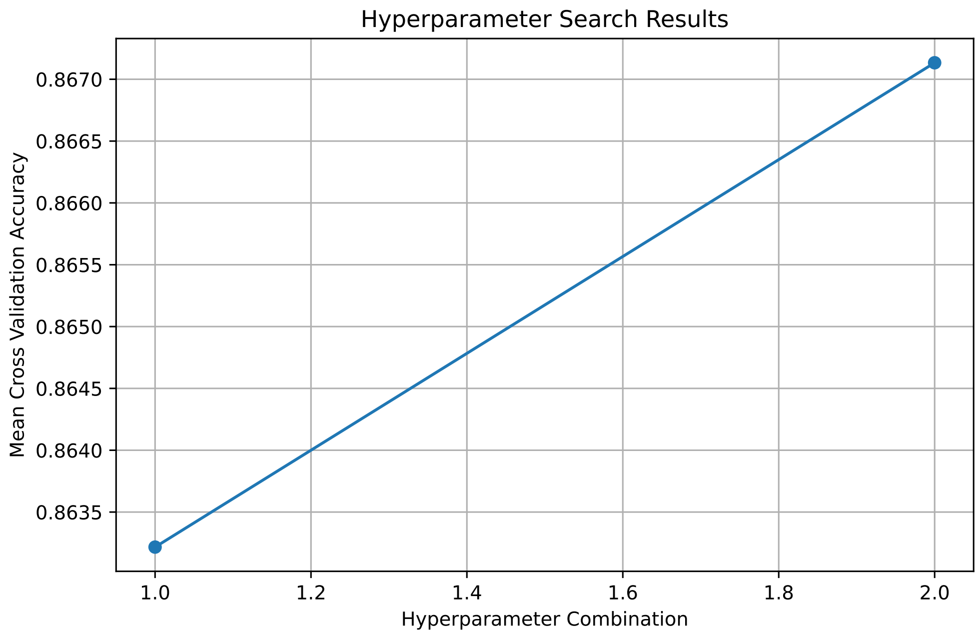

Figure 7: Hyperparameter Search Results

The hyperparameter search results illustrate the variation in cross-validation accuracy for different parameter combinations. The model with two hidden layers achieved the highest cross-validation accuracy of 86.71%, making it the optimal configuration selected for retraining.

0.1 Baseline vs Optimized Model Comparison

The optimized model was compared with the baseline MLP using standard performance metrics to evaluate the effect of hyperparameter tuning.

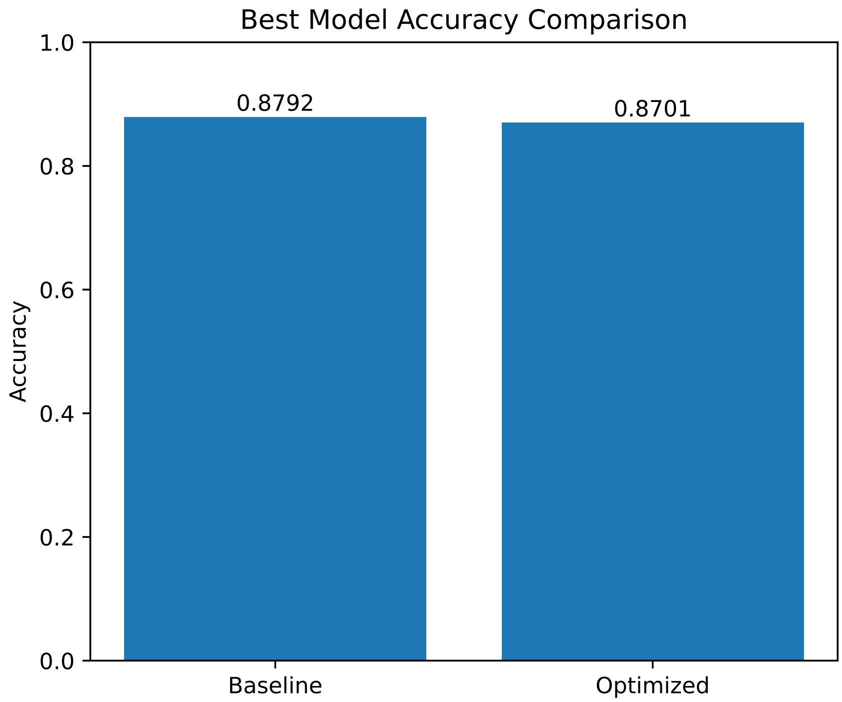

Figure 8: Baseline vs Optimized Model Accuracy Comparison

The comparison shows that the baseline model achieved slightly higher testing accuracy than the optimized model. Although hyperparameter tuning reduced the training time significantly, it did not improve the final test performance, indicating that the baseline configuration generalized better for this experiment.

Best Hyperparameters

Table 8: Best Hyperparameters Obtained from Randomized Search

Hyperparameter	**Best Value**
Optimizer	Adam
Learning Rate	0.001
Hidden Layers	2
Hidden Neurons	64
Activation Function	ReLU
Dropout Rate	0.0
Epochs	5
Batch Size	64
Cross-validation Accuracy	0.8671
Testing Accuracy	0.8701

Table 9: Performance Comparison Between Baseline and Optimized Models

Metric	**Baseline**	**Optimized**
Accuracy	0.8792	0.8701
Precision	0.8814	0.8718
Recall	0.8792	0.8701
F1-score	0.8783	0.8660
Training Time (s)	145	19

8. Discussion

1. **Which optimization method was used?**
   Randomized Search Cross Validation (RandomizedSearchCV) was used to find the best hyperparameters.
2. **Which hyperparameters were selected?**
   The best model used 2 hidden layers, 64 neurons, ReLU activation, Adam optimizer, learning rate of 0.001, batch size of 64, no dropout, and 5 epochs.
3. **Did optimization improve performance?**
   No. The optimized model achieved a slightly lower testing accuracy than the baseline model.
4. **Which hyperparameter had the greatest impact?**
   The number of hidden layers had the greatest impact on the model's performance.
5. **Compare Grid Search and Randomized Search.**
   Grid Search checks all parameter combinations, while Randomized Search checks only a few randomly selected combinations. Randomized Search is faster.
6. **Recommend the best MLP configuration.**
   The recommended configuration is 2 hidden layers, 64 neurons, ReLU activation, Adam optimizer, learning rate 0.001, batch size 64, no dropout, and 5 epochs.

9. Conclusion

In this experiment, a Multi-Layer Perceptron (MLP) was successfully implemented for Fashion-MNIST image classification. Data preprocessing, model training, performance evaluation, and hyperparameter optimization using Randomized Search Cross Validation were carried out systematically. The baseline model achieved a testing accuracy of 87.92%, while the optimized model achieved 87.01%. Although hyperparameter optimization reduced the training time, it did not improve the final testing accuracy, demonstrating that careful selection of network architecture and training parameters is essential for achieving optimal performance.

10. References

1. Goodfellow et al., *Deep Learning.*
2. Bishop, *Pattern Recognition and Machine Learning.*
3. Haykin, *Neural Networks and Learning Machines.*
4. Fashion-MNIST Dataset.
5. TensorFlow/Keras Documentation.